GPS DoP and iterative positioning

Oscar Gonzalo Aran
oscar.gonzalo@estudiantat.upc.edu
EETAC – Universitat Politècnica de Catalunya (UPC)
Castelldefels, Spain

Adam Ruiz Herrero
adam.ruiz@estudiantat.upc.edu
EETAC – Universitat Politècnica de Catalunya (UPC)
Castelldefels, Spain

***Abstract*—This report studies how the geometry of the visible GPS satellites affects the accuracy of the position computed by a receiver, and how the navigation system of equations is solved. In the first part, the Dilution of Precision (DoP) at EETAC (Castelldefels) is computed for six hours starting at 12:00 UTC on 14/10/2026, using the almanac published by CelesTrak and a 10° elevation mask. At the initial epoch, 9 satellites are visible and GDOP = 2.35, PDOP = 2.02, HDOP = 1.05 and VDOP = 1.73. The curves agree with those of the online tool GNSS Mission Planning to within 0.05 at the nodes of the tool. In the second part, the receiver position is obtained from the ten pseudoranges assigned to our group by successive linearisations and least squares. The iteration reaches a displacement of less than 10 m after five iterations and converges to latitude -75.9284°, longitude -20.6358° and height 4.0 m.**

***Index Terms*—GPS, dilution of precision, DoP, least squares, pseudorange, navigation equations.**

I. Introduction

The accuracy of a GPS position has two origins. On the one hand, the measured pseudoranges contain errors that cannot be fully cancelled (ionosphere, relativistic effects, etc.). On the other hand, finding the position of the user means solving a system of equations, and depending on the condition of its matrix the solution may show an error larger than that of the pseudoranges. This amplification is called Dilution of Precision (DoP), and it depends only on the relative positions of the satellites in line of sight (LoS) with respect to the user [1].

This fourth session builds on the previous ones. The functions that give the ECEF position of every satellite (`kepler2ecef.m`) and the LoS test of Lab. Session 3 (`lla2xyz.m`, `ecef2ned.m`, `ned2spherical.m`) are reused. Four functions (`dop.m`, `qmat.m`, `q2dop.m` and `reference.m`) and two main programs (`main5.m` and `main6.m`) have been written. Section II explains the method, and Section III presents the results and the comparison with an online tool.

II. Method

The constants are the same as in the previous sessions: $G = 6.67384 \cdot 10^{-11}$ m^{3}kg^{-1}s^{-2}, $M_e = 5.972 \cdot 10^{24}$ kg, $\omega_e = 7.2921151467 \cdot 10^{-5}$ rad/s and a tolerance of 10^{-8} for the Kepler equation. The ECEF position $\mathbf{r}_S$ of a satellite at the time t is obtained from the almanac with `kepler2ecef.m`, with the right ascension at the time of applicability $\Omega_0 = \Omega_{week} - \omega_e\, t_{oa}$ and $\Delta t = t - t_{oa}$.

A. Navigation equations and DoP

The linearised navigation equation is $\Delta\boldsymbol{\rho} = \mathbf{H}\,\Delta\mathbf{u}$, where $\Delta\mathbf{u}$ contains the three components of the displacement of the user and the clock term. If the measured pseudoranges have an unknown error vector $\boldsymbol{\eta}$, then $\Delta\boldsymbol{\rho}' = \Delta\boldsymbol{\rho} + \boldsymbol{\eta}$ and the least squares (LS) solution is

$$\Delta\mathbf{u}' = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\Delta\boldsymbol{\rho}'. \tag{1}$$

The error of the position is therefore

$$\boldsymbol{\varepsilon} = \Delta\mathbf{u}' - \Delta\mathbf{u} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\boldsymbol{\eta}, \tag{2}$$

so the pseudorange error is amplified by a factor that depends only on $\mathbf{H}$. If the components of $\boldsymbol{\eta}$ are independent, with zero mean and the same variance σ^2, the covariance matrix of $\boldsymbol{\varepsilon}$ is

$$E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T] = \sigma^2\mathbf{Q}, \qquad \mathbf{Q} = (\mathbf{H}^T\mathbf{H})^{-1}, \tag{3}$$

which is always a 4×4 matrix, however many satellites are used. Since $\boldsymbol{\varepsilon}$ is expressed in ECEF coordinates, it is rotated to the local NED frame with $\boldsymbol{\delta} = \mathbf{R}\boldsymbol{\varepsilon}$, where

$$\mathbf{R} = \begin{bmatrix} -\sin\varphi\cos\lambda & -\sin\varphi\sin\lambda & \cos\varphi & 0 \\ -\sin\lambda & \cos\lambda & 0 & 0 \\ -\cos\varphi\cos\lambda & -\cos\varphi\sin\lambda & -\sin\varphi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4}$$

is the ECEF-to-NED matrix of the previous session with an additional row and column that leave the time component unchanged. The covariance of $\boldsymbol{\delta}$ is $\sigma^2\mathbf{Q}'$ with

$$\mathbf{Q}' = \mathbf{R}\,\mathbf{Q}\,\mathbf{R}^T = \begin{bmatrix} q_{NN} & q_{NE} & q_{ND} & q_{Nct} \\ q_{EN} & q_{EE} & q_{ED} & q_{Ect} \\ q_{DN} & q_{DE} & q_{DD} & q_{Dct} \\ q_{ctN} & q_{ctE} & q_{ctD} & q_{ctct} \end{bmatrix}. \tag{5}$$

Only the diagonal elements are needed:

$$\mathrm{HDOP} = \sqrt{q_{NN} + q_{EE}}, \qquad \mathrm{VDOP} = \sqrt{q_{DD}}, \tag{6}$$

$$\mathrm{PDOP} = \sqrt{q_{NN} + q_{EE} + q_{DD}}, \qquad \mathrm{TDOP} = \sqrt{q_{ctct}}, \tag{7}$$

$$\mathrm{GDOP} = \sqrt{\mathrm{PDOP}^2 + \mathrm{TDOP}^2}. \tag{8}$$

B. Step 1: system matrix (`dop.m`)

The function has the prototype `[gdop,pdop,hdop,vdop,tdop] = dop(Oe,G,Me,tol,esec,ulla,NS,Eph)`. It starts from the program of the previous session, without the call to `viewsat`. First the user position `ulla` (latitude and

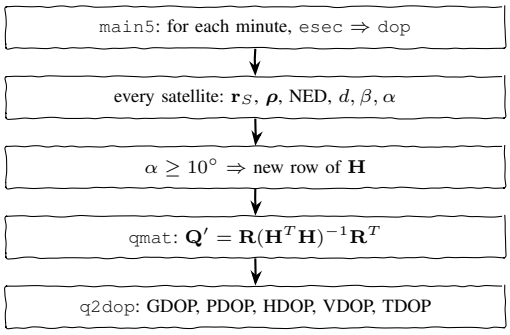


Fig. 1. Computation chain of `main5.m`.

longitude in degrees, height in metres) is converted to ECEF with `lla2xyz.m`. Then a loop over the N_S satellites of the almanac computes, for each one, $\mathbf{r}_S$ with `kepler2ecef.m`, the pointing vector $\boldsymbol{\rho} = \mathbf{r}_S - \mathbf{r}_U$, its NED components with `ecef2ned.m` and the distance d, azimuth and elevation α with `ned2spherical.m`. If $\alpha \geq 10^\circ$, the satellite is in LoS and a new row is added to $\mathbf{H}$:

$$\mathbf{H}_i = \begin{bmatrix} \dfrac{\rho_x}{d} & \dfrac{\rho_y}{d} & \dfrac{\rho_z}{d} & 1 \end{bmatrix}. \tag{9}$$

The first three entries are the direction cosines of the line of sight and the last one corresponds to the receiver clock. Finally, `dop.m` has as many rows in $\mathbf{H}$ as satellites in LoS at the time `esec`, and calls `qmat.m` and `q2dop.m`. If the global variable `fd1` exists, the function writes to `main5.dat` the PRN, azimuth and elevation of the visible satellites. The sign convention of the first three columns of $\mathbf{H}$ does not change the DoP: changing it multiplies $\mathbf{H}$ on the right by a diagonal matrix $\mathbf{S}$ with entries ± 1, so that $\mathbf{Q} \to \mathbf{SQS}$, and the diagonal of $\mathbf{Q}'$ is unchanged.

C. Step 2: matrix $\mathbf{Q}'$ (`qmat.m`)

The function `Q = qmat(H,ulat,ulong)` computes $\mathbf{Q} = (\mathbf{H}^T\mathbf{H})^{-1}$ with `inv`, builds $\mathbf{R}$ from the latitude and longitude of the user (in radians) and returns $\mathbf{Q}' = \mathbf{RQR}^T$. The matrix is also written to `main5.dat`.

D. Step 3: DoP values (`q2dop.m`)

The function `[gdop,pdop,hdop,vdop,tdop] = q2dop(Q)` takes the diagonal of $\mathbf{Q}'$ and evaluates the five expressions of the previous subsection.

E. Main program (`main5.m`)

The program sets the epoch (14/10/2026, 12:00:00 UTC, an exact hour to compare with the online tool), reads the almanac with `almanac.m` and sets the user position to EETAC, $\varphi = 41.275556^\circ$ N, $\lambda = 1.986944^\circ$ E, $h = 4$ m. It then calls `dop.m` every minute during the next six hours (361 evaluations). The vector `tm` contains the elapsed time in minutes after the initial epoch and the vectors `gdop`, `pdop`, `hdop`, `vdop` and `tdop` the five DoP values; they are plotted at the end. The chain executed by the program is shown in Fig. 1.

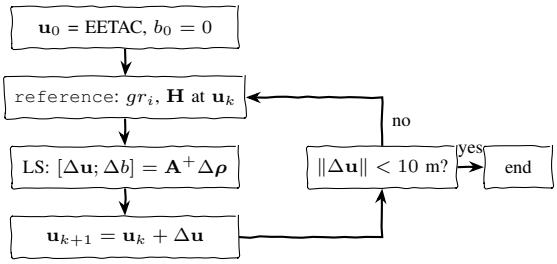


Fig. 2. Iterative solution of the navigation equations.

F. Step 4: reference position (`reference.m`)

To solve the system for the actual position of the user, the linearisation is made around a reference position, the last known position of the user. The function `reference.m`, with prototype

```
[gr,H] = reference(Oe,G,Me,tol,esec,refecef,NS,Eph,satlist)
```

is a modification of `dop.m`. The position `refecef` is given in ECEF coordinates, there is no LoS test, and the satellites are those of the vector `satlist`, whether or not they are visible from the reference position. For each satellite i it computes the vector $\mathbf{gr}_i = \mathbf{r}_{S,i} - \mathbf{r}_{ref}$, its magnitude gr_i (geometric range) and the row $\mathbf{H}_i = [gr_x/gr_i,\ gr_y/gr_i,\ gr_z/gr_i,\ 1]$. It returns the vector `gr` and the matrix $\mathbf{H}$, which are also written to `main6.dat`.

G. Step 5: iterative solution (`main6.m`)

The pseudorange of satellite i is $pr_i = \|\mathbf{r}_{S,i} - \mathbf{u}\| + c\,\Delta t_u$, where $\mathbf{u}$ is the unknown position and $c\,\Delta t_u$ is the clock term, in metres. Let $\mathbf{u}_k$ be the position of iteration k and b_k the clock term (initially $\mathbf{u}_0$ = EETAC and $b_0 = 0$). The first-order expansion is

$$pr_i - gr_i - b_k = -\begin{bmatrix} \dfrac{gr_x}{gr_i} & \dfrac{gr_y}{gr_i} & \dfrac{gr_z}{gr_i} \end{bmatrix} \Delta\mathbf{u} + \Delta b, \tag{10}$$

where gr_i is the geometric range from $\mathbf{u}_k$. In matrix form $\Delta\boldsymbol{\rho} = \mathbf{A}\,[\Delta\mathbf{u}; \Delta b]$, with $\Delta\rho_i = pr_i - gr_i - b_k$ and $\mathbf{A} = [-\mathbf{H}_{(:,1:3)}\ \ \mathbf{1}]$. The system is overdetermined (ten satellites), so it is solved by LS with the pseudoinverse (`pinv` in MATLAB):

$$[\Delta\mathbf{u}; \Delta b] = \mathbf{A}^+\Delta\boldsymbol{\rho},\ \ \mathbf{u}_{k+1} = \mathbf{u}_k + \Delta\mathbf{u},\ \ b_{k+1} = b_k + \Delta b. \tag{11}$$

The measured pseudoranges pr_i do not change along the iterations, whereas gr_i and $\mathbf{H}$ are recomputed with `reference.m` at each iteration (Fig. 2). The loop ends when $\|\Delta\mathbf{u}\| < 10$ m. At every iteration, the ECEF position is converted to latitude, longitude and height with `ecef2lla.m` and stored as a row of the matrix `computedlla`. At the end, the intermediate positions are drawn on the world map with `plotworld.m`.

H. Debugging files

`main5.dat` contains, for each of the 361 epochs, the list of visible satellites, the matrix $\mathbf{Q}'$ and the five DoP values. `main6.dat` contains the matrix $\mathbf{H}$ of each iteration, the number of iterations and the computed position.

TABLE I
DoP COMPUTED BY `main5.m` / READ FROM GNSS MISSION PLANNING.

t [min]	GDOP	PDOP	HDOP	VDOP
0	2.35 / 2.36	2.02 / 2.03	1.05 / 1.05	1.73 / 1.74
10	2.72 / 2.71	2.30 / 2.29	1.09 / 1.08	2.03 / 2.02
20	2.00 / 2.00	1.74 / 1.73	0.93 / 0.93	1.47 / 1.47
30	1.70 / 1.70	1.50 / 1.50	0.84 / 0.84	1.24 / 1.24
60	1.69 / 1.69	1.51 / 1.50	0.84 / 0.84	1.25 / 1.25
120	2.17 / 2.14	1.90 / 1.88	0.89 / 0.88	1.68 / 1.66
180	1.77 / 1.77	1.56 / 1.57	0.89 / 0.88	1.29 / 1.29
240	1.76 / 1.76	1.56 / 1.55	0.96 / 0.96	1.22 / 1.22
300	2.05 / 2.03	1.78 / 1.77	0.96 / 0.96	1.50 / 1.49
360	2.86 / 2.85	2.45 / 2.44	1.38 / 1.37	2.02 / 2.01

III. RESULTS

A. DoP during six hours

The DoP was computed at EETAC from 12:00 to 18:00 UTC on 14/10/2026 with a time step of one minute, using the almanac of GPS week 392 (ToA = 319488 s) from CelesTrak [2] and a 10° mask. At the initial epoch nine satellites are visible (PRN 5, 14, 15, 18, 20, 21, 22, 24 and 30), the same ones as in the previous session, and `main5.dat` gives

$$\mathbf{Q}' = \begin{bmatrix} 0.618 & 0.007 & -0.089 & -0.048 \\ 0.007 & 0.477 & 0.671 & 0.479 \\ -0.089 & 0.671 & 2.979 & 1.977 \\ -0.048 & 0.479 & 1.977 & 1.427 \end{bmatrix},$$

which leads to GDOP = 2.345, PDOP = 2.018, HDOP = 1.047, VDOP = 1.726 and TDOP = 1.194.

Fig. 3 shows the result next to the capture of GNSS Mission Planning [3], configured with the same date, six hours, GPS only, a cutoff of 10° and the position of EETAC (41.27557°, 1.98725°, height 0 m). The shapes of the curves are the same, and Table I lists both sets of values at several nodes of the tool, which computes a point every ten minutes. To read the tool, the four curves of the capture were located pixel by pixel and converted with the axis scales. The differences are of at most 0.05 over all 37 nodes (root mean square 0.014 for GDOP, 0.012 for PDOP, 0.004 for HDOP and 0.011 for VDOP).

The curves of `main5.m` are not smooth because the number and the geometry of the satellites in LoS change along the six hours (the tool, with a point every ten minutes, joins its nodes with straight segments). From `main5.dat`, between 8 and 12 satellites are visible. The first sudden increase is at minute 7, when PRN 18 goes below the mask (8 satellites left): GDOP goes up to 2.72 at minute 10 and decreases when PRN 23 (minute 16) and PRN 17 (minute 21) rise. The maximum GDOP, 2.87, appears at minute 359, when PRN 24 sets and only 8 satellites remain, and the minimum, 1.44, at minute 201 with 11 satellites well distributed in the sky. HDOP is the smallest of the factors during the six hours (between 0.7 and 1.4), because satellites are seen all around the horizon, whereas VDOP is always larger because all satellites are above the user, and none of them gives information from below. TDOP is not provided by the online tool; in our results it is between 0.6 and 1.5.

TABLE II
PSEUDORANGES OF GROUP 18 (05/10/2026 19:32:59 UTC).

PRN	pr [m]	PRN	pr [m]
2	23539247.564	20	24287292.281
8	22210360.932	22	23648471.354
10	20589492.405	23	20904026.072
14	22962052.341	24	23304295.251
15	22685456.754	27	23053381.928

TABLE III
ITERATIONS OF `main6.m`.

k	$\|\Delta\mathbf{u}\|$ [m]	φ [°]	λ [°]	h [m]
1	14335641.94	−85.188183	−74.785460	3040121.48
2	3304853.52	−76.857042	−22.163342	88481.43
3	142383.45	−75.930847	−20.639599	91.94
4	304.29	−75.928405	−20.635823	4.00
5	0.00	−75.928405	−20.635823	4.00

B. Iterative solution

The file `Group_18.dat` gives the epoch 05/10/2026 19:32:59 UTC and the pseudoranges of the ten satellites in LoS from the receiver, which is placed at a random position of the globe, at a height of 4 m over the ellipsoid (Table II). The almanac of week 391 with ToA = 233472 s was used.

Table III shows the evolution of the iteration, starting at EETAC. The displacement decreases very quickly, as expected for a linearisation that approaches the solution: the last displacement larger than 10 m (304 m, iteration 4) is followed by one of a few millimetres, so the loop ends in iteration 5. Already in iteration 4 the height is 4.00 m, the value that indicates that the solution is correct. The final result is

$$\varphi = -75.9284^\circ, \quad \lambda = -20.6358^\circ, \quad h = 3.9999 \text{ m},$$

that is, a point in Antarctica. The clock term obtained is below 10^{-3} m, as the pseudoranges do not contain any clock error, and the residuals $pr_i - gr_i$ of the ten satellites are all smaller than 10^{-4} m, so the ten equations are consistent with a single position. Fig. 4 shows the intermediate positions on the world map, and the displacement of each iteration.

IV. CONCLUSIONS

- The DoP depends only on the matrix of the navigation equations, that is, on the geometry of the satellites in LoS. It is obtained from the diagonal of $\mathbf{Q}' = \mathbf{R}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{R}^T$, so that HDOP and VDOP are referred to the local horizontal plane and vertical of the user.
- At EETAC, on 14/10/2026, the GDOP stays between 1.44 and 2.87 during six hours, with 8 to 12 satellites visible. The sudden changes of the curves are due to satellites that rise above or go below the 10° mask.
- The results of `main5.m` agree with GNSS Mission Planning to within 0.05 at the nodes of the tool, with a time step of one minute instead of ten.

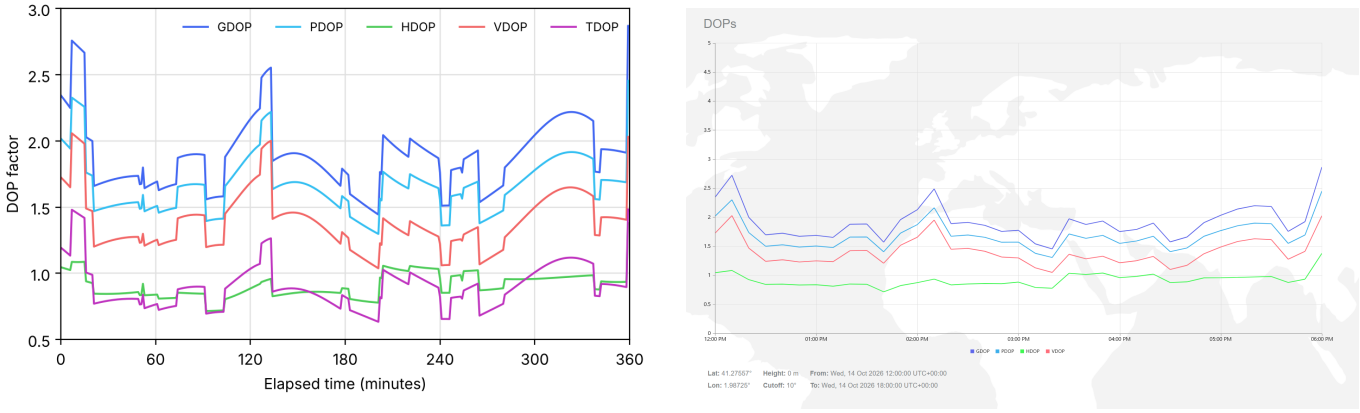


Fig. 3. DoP at EETAC from 12:00 to 18:00 UTC on 14/10/2026. Left: result of `main5.m` (step of one minute). Right: capture of GNSS Mission Planning [3] (GPS only, cutoff 10°, UTC time zone; the tool does not give TDOP).

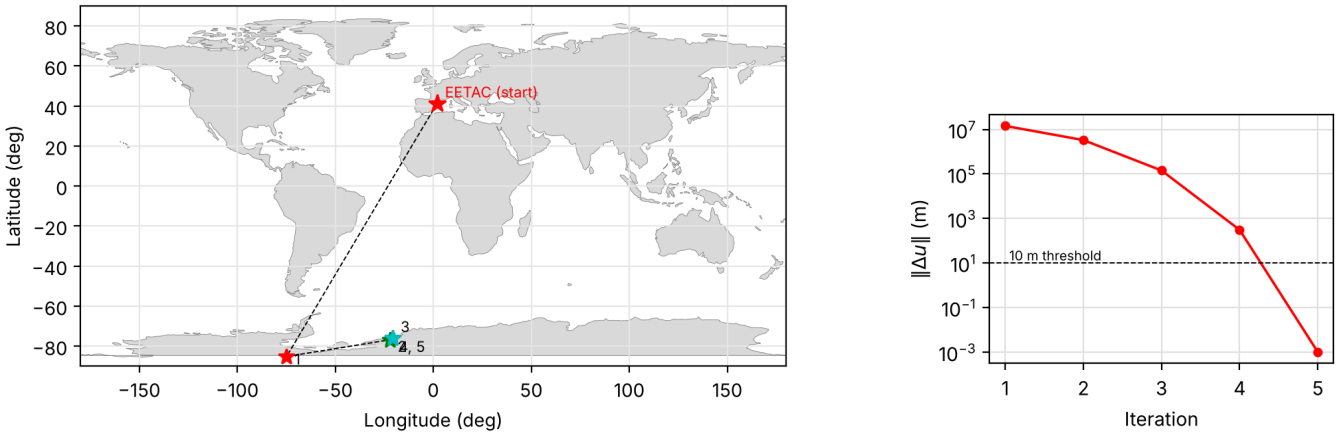


Fig. 4. Left: intermediate positions of the iterative solution on the world map (the red star is EETAC, the initial reference position; iterations 4 and 5 coincide). Right: magnitude of the displacement $\|\Delta\mathbf{u}\|$ at each iteration.

- The navigation equations are solved by linearisation around a reference position and LS with the pseudoinverse. Starting from EETAC, which is on the other side of the globe, five iterations give a displacement smaller than 10 m and the expected height of 4 m.

References

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